Project Title: House Price Prediction Using Machine Learning

Project Overview

The objective of this project is to build a machine learning model that predicts house prices based on property features such as area, number of bedrooms, bathrooms, stories, location-related attributes, and other facilities. The project helps in estimating fair property values and supports data-driven decisions in the real estate sector.

Dataset Description

The project uses the Housing Prices Dataset containing information about different residential properties. The dataset includes numerical features like area, bedrooms, bathrooms, parking, and categorical features such as furnishing status, main road availability, air conditioning, and preferred location. The target variable for prediction is the house price.

Data Preprocessing

The dataset was loaded and explored using Pandas. Data quality checks were performed to identify missing values and duplicate records. Duplicate rows were removed, and categorical variables were converted into numerical format using one-hot encoding. The cleaned dataset was then prepared for machine learning model training.

Machine Learning Models

Two regression algorithms were implemented:

1. Linear Regression — Used as a basic regression model to understand the relationship between property features and prices.
2. Random Forest Regressor — Used to capture complex patterns and improve prediction performance.

The dataset was divided into training and testing sets using an 80:20 split. Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

Results and Insights

The Random Forest Regressor provided better performance compared to Linear Regression because it can handle complex relationships between different property features. The analysis showed that features such as area, number of bathrooms, bedrooms, and location-related factors have a significant influence on house prices.

The visual analysis included a house price distribution histogram, correlation heatmap, and actual versus predicted price comparison chart to understand patterns in the data.

Business Recommendation

Real estate businesses can use this machine learning model to estimate property prices accurately, reduce dependency on manual comparisons, and support better pricing decisions. Understanding important features affecting prices can help sellers and buyers make informed decisions.

Conclusion

This project demonstrates how machine learning regression techniques can be applied to real estate data for price prediction. By combining data preprocessing, model training, evaluation, and visualization, the system provides useful insights for property valuation and future real estate analysis.